# PHY · Relativity — syllabus points and past-paper evidence

Every award of the 2016–2025 papers, labelled to the 2026/2027 Year 12 syllabus content (identical in both). Statistics from the labels; notes written by AI from that evidence: review before use.

## Pattern 1: Postulates of special relativity

Syllabus: S26-U3-SU-15 / S27-U3-SU-15

Frequency: 3 of 10 exam years; 9 marks in all.

Marks reward: Recall 3, Final answer 2, Qualitative explanation 2, Select the principle or relationship 1, Substitution and working 1

Past questions: 2016R-4(b)；2017R-5(d)；2017R-7；2023P-14(aii)；2023R-9

Summary: This point tests the two postulates of special relativity: the speed of light in a vacuum is the same for every observer, and the laws of physics take the same form in every inertial frame. It usually appears as short recall of the postulates, as multiple-choice or short-answer questions about what an observer measures for light from different sources, and as reasoning questions where a moving observer's measurements of light travel are linked to a stationary observer's.

Recognise: Look for questions that mention inertial frames, observers moving at constant velocity, light from a moving or distant source, or a signal sent between two points measured by two observers. Questions asking 'on which two postulates is the theory based' or 'why does one observer's measured speed equal another's' are testing this point directly. Any question where the answer depends on whether the light speed changes with source, colour, or observer motion is a sign the postulates are needed.

Steps

1. State the postulates in your own words: light in a vacuum has the same speed c for every observer, and physical laws are identical in all inertial frames.
2. For a light signal, identify the distance travelled and the time taken in each frame separately, using the observer's own measurements.
3. Apply speed = distance / time in each frame: c = d/t for the stationary observer and c = d'/t' for the moving observer.
4. Because both observers measure the same c, equate the two expressions and rearrange for the unknown quantity.
5. For questions about the source or colour of light, conclude that the measured speed is independent of the emitter's motion, so the speed is the same for all cases.
6. For symmetry questions about two observers moving relative to each other, argue that no frame is preferred, so the magnitude of relative velocity must be the same when measured from either side, with only the direction reversed.

Where the marks are

- Full marks for KNOW require both postulates stated clearly, with the constant light speed and the equivalence of inertial frames each given separately.
- A mark for ANSWER is given for choosing the option that light speed is independent of source or colour, not for guessing based on wavelength.
- A mark for PRINCIPLE is given for writing that each observer's measured speed equals c, linking the postulate to the measurement.
- A mark for CALC is given for correctly rearranging the speed relationship to make the unknown the subject.
- EXPLAIN marks are given for citing the absence of a preferred frame and the symmetry argument, each as its own justified statement.

Pitfalls

- Writing 'light speed is constant' without saying it is constant for all observers, which loses the point

about frames of reference.

- Choosing an answer that says light from a moving source is faster or slower, because the source speed does not change the measured value of c.
- Confusing a stationary and moving frame and using the same distance or time for both observers, instead of each observer's own measurement.
- Stating that a 'preferred frame' or 'ether' exists, which contradicts the second postulate.
- Giving a reason for equal velocity magnitudes without naming symmetry or the lack of a preferred frame, so the explanation is incomplete.

## Pattern 2: High speeds beyond Newtonian physics

Syllabus: S26-U3-SU-16 / S27-U3-SU-16

Frequency: 1 of 10 exam years; 1 marks in all.

Marks reward: Graph work 1

Past questions: 2025R-10

Summary: This point asks why Newtonian mechanics fails for objects moving at very high speeds, with the usual focus on relativistic kinetic energy and what happens as speed approaches the speed of light c. It is rarely examined on its own: the sampled past papers contain only one part, a short explain question that uses a graph of kinetic energy against speed together with a formula from the Formulae and Data booklet.

Recognise: Look for a graph or table of kinetic energy (or momentum or mass) against speed, a request to explain why an object with mass cannot reach c, or a comparison between the Newtonian formula and the observed behaviour at high speed. The word 'explain' with a reference to a graph or formula is the usual signal.

Steps

1. Recall that the Newtonian kinetic energy formula $E_k = 1/2\ mv^2$ predicts a finite energy at every speed, including speeds near c.
2. Recall the relativistic kinetic energy $E_k = (\gamma - 1)mc^2$, where $\gamma = 1 / \sqrt{1 - v^2/c^2}$, from the Formulae and Data booklet.
3. Examine the behaviour as v approaches c: the term $1 - v^2/c^2$ approaches 0, so $\gamma$ approaches infinity and $E_k$ grows without limit.
4. Read the graph: the curve rises steeply and becomes asymptotic to the vertical line at $v = c$, so $E_k$ tends to infinity as v tends to c.
5. Conclude that an object with mass would need infinite energy to reach c, which is not physically possible, so no massive object can attain the speed of light.
6. State that the Newtonian model is inadequate at high speed because it gives a finite energy at c, which disagrees with observation.

Where the marks are

- One mark for stating that $E_k$ tends to infinity (the graph is asymptotic to $v = c$) as speed approaches c.
- Further marks usually go for linking the infinite energy to the impossibility of reaching c, using the relativistic formula or the gamma factor.
- Marks are given for naming the relativistic formula from the booklet, not for a bare assertion that the speed limit is c.
- Examiners expect the conclusion to be stated as a consequence of the graph or formula, not left as a description.

Pitfalls

- Saying the energy becomes large or very large rather than stating that it tends to infinity, which loses the key point.
- Describing the graph without connecting the asymptote to the speed of light.

- Quoting the Newtonian formula as the reason, when the point is that it fails at high speed.
- Claiming the object is 'too heavy' to accelerate, instead of explaining the infinite-energy limit.
- Forgetting to state the conclusion, that no object with mass can attain c.

## Pattern 3: Relativity of simultaneity

Syllabus: S26-U3-SU-17 / S27-U3-SU-17

Frequency: 2 of 10 exam years; 6 marks in all.

Marks reward: Qualitative explanation 6

Past questions: 2019R-11(c); 2022R-11(c)

Summary: This point examines whether two events judged simultaneous in one inertial frame are also simultaneous in another. In past papers it appears as a short EXPLAIN question, usually asking how two observers can get apparently contradictory length measurements of a moving object, with both answers resting on the fact that 'at the same time' depends on the observer.

Recognise: Two inertial observers, one moving relative to the other, and a measurement that needs the positions of two ends (or two events) at the same moment, such as whether a moving object fits between two markers, or why two lengths disagree. Cue words: simultaneously, at the same time, contradictory results, both be correct.

Steps

1. Identify the measurement procedure. A length is found by noting the positions of both ends at the same instant in the observer's own frame.
2. Identify the frame in which the two events are simultaneous (delta t = 0 in that frame) and the separation delta x between the two events.
3. Apply the Lorentz transformation for time to the second frame moving at speed v: delta t' = gamma(delta t - v delta x / c^2), where gamma = 1 / sqrt(1 - v^2/c^2).
4. Show that delta t' is not zero whenever delta x is not zero, so the events are not simultaneous in the other frame. The event further along the direction of motion is timed earlier in the moving frame.
5. Connect this to length: each observer takes the ends at a different pair of times, so the moving object's length is L = L0 / gamma for the moving observer and the other observer's 'fitting' result changes accordingly.
6. Conclude that both results are correct within their own frames. The discrepancy comes from different definitions of simultaneity, not from an error in measurement.

Where the marks are

- A mark for stating that measuring a length requires both ends to be located at the same time, in the observer's own frame.
- A mark for stating that simultaneity is frame-dependent, so each observer's 'same time' is different.
- A mark for linking the non-simultaneity to the different lengths obtained, with length contraction (L = L0 / gamma) or the transformed time difference.
- A mark for a clear conclusion that both observers are correct within their own frame of reference.

Pitfalls

- Saying the events happen at different times in reality, rather than being judged simultaneous in one frame and not in another.
- Explaining the difference with time dilation alone, when the key idea is the non-simultaneity of the measurements.
- Stating length contraction without explaining why the measurement procedure depends on simultaneity.
- Claiming one observer made a mistake, or that one measurement is 'the real one'. Both are valid in their own frames.
- Using everyday absolute time, where simultaneity is the same for everyone, and so missing the central

point of the answer.

## Pattern 4: Time dilation and length contraction

Syllabus: S26-U3-SU-18 / S27-U3-SU-18

Frequency: 10 of 10 exam years; 93 marks in all.

Marks reward: Select the principle or relationship 26, Final answer 23, Substitution and working 21, Qualitative explanation 13, Units, significant figures, direction 5, Recall 2, Graph work 2, Evaluate and conclude 1

Past questions: 2016R-12(a); 2016R-12(b); 2016R-7; 2017P-17(bi); 2017P-17(bii); 2017P-17(biii); 2017R-2; 2018P-16(a); 2018P-16(b); 2018P-16(c); 2018P-16(d); 2019P-15(d); 2019R-11(a); 2019R-11(b); 2019R-11(c); 2019R-9; 2020P-14(ci); 2020P-14(cii); 2020P-14(d); 2020P-14(e); 2020P-15(a); 2020P-15(b); 2020P-15(c); 2021P-18(c)...

Summary: This point tests whether students can treat time and length as observer-dependent: a moving clock runs slow and a moving object is shortened as seen from a frame where it is not at rest, and the relativistic velocity-addition rule combines speeds. It usually appears as multi-part calculations that chain a velocity or time result into a distance or travel-time answer, plus short explanations of why an observer sees what they see.

Recognise: Look for two frames of reference (Earth and a spacecraft, a muon and an Earth observer, two spacecraft moving toward or away from each other), speeds given as fractions of c, a 'proper' or rest-frame time or length given in one frame and asked for in another, or wording such as 'as measured by', 'as observed from', 'in the frame of'. Questions asking to explain why a particle or clock behaves differently from the classical prediction also belong here.

### Steps

1. Name the frame for each quantity: the proper time t0 or proper length l0 is measured in the frame where the clock or object is at rest; the quantity asked for is measured in the other frame.
2. For time, use t = t0 / sqrt(1 - v^2/c^2) so the observer who sees the moving clock measures a longer interval; for length, use l = l0 sqrt(1 - v^2/c^2) so the moving object appears shorter along its direction of motion.
3. For two speeds in the same line, use u = (u' + v)/(1 + u'v/c^2), where u' is the velocity in the moving frame and v is the speed of that frame relative to the observer; choose one positive direction and assign signs consistently.
4. Keep the Lorentz factor 1/sqrt(1 - v^2/c^2) with extra significant figures until the final answer, then round to the precision the question asks for.
5. If the result feeds a later part, carry it through in the correct frame, for example speed times time in the Earth frame, and check the answer is below c.
6. For explanation parts, state which quantity is dilated or contracted, which observer measures it, and link that to the observed outcome (e.g. a particle surviving longer or travelling a shorter distance).

### Where the marks are

- Most marks go to naming and applying the correct relativistic relationship before substituting numbers.
- A mark is usually given for consistent sign convention and for identifying which frame the unknown belongs to.
- Calculation marks depend on substituting the right values in the right places, including the square-root factor.
- The final answer needs correct units and sensible rounding; a correct method with a wrong final value still earns most method marks.
- Explanation marks reward naming the effect (time dilation or length contraction), stating the observer, and making the link to the observed result.

### Pitfalls

- Putting the proper time or proper length in the wrong place, such as using the moving-frame value as t0 or l0.
- Using ordinary addition of velocities or ignoring the denominator term in the velocity-addition formula.
- Sign errors when objects move in opposite directions, which change the sign of the result.
- Explanations that say time 'slows' or length 'shrinks' without saying whose measurement it is, or that skip the simultaneity point when measuring a moving length.
- Unit and rounding slips: mixing km with m, seconds with years, or rounding the Lorentz factor too early.

## Pattern 5: Muons as evidence for special relativity

Syllabus: S26-U3-SHE-02 / S27-U3-SHE-02

Frequency: 1 of 10 exam years; 2 marks in all.

Marks reward: Qualitative explanation 2

Past questions: 2016R-12(b)

Summary: This point asks you to use high-speed muons from the upper atmosphere as experimental evidence for special relativity. It usually appears as an explain question: why more muons reach the ground than a classical calculation predicts. The point has been examined rarely, with one matching part in the set reviewed.

Recognise: Look for a question about muons (or another short-lived, fast particle) created high in the atmosphere that reach the surface, a comparison with a classical prediction, and a request to explain the difference. Key words: high speed, lifetime, distance travelled, observer on Earth, muon's viewpoint.

Steps

1. State the classical prediction: a muon at rest has a short mean lifetime, so at near light speed it travels only a few hundred metres before decaying, far less than the atmosphere's depth.
2. Recognise that the muon's speed is close to c, so relativistic effects are significant. Use gamma = 1/ sqrt(1 - v^2/c^2).
3. Earth-observer frame: the muon's lifetime is dilated, so its observed lifetime is gamma times its proper lifetime, and it travels a distance v x gamma x tau.
4. Muon frame: the atmosphere's thickness is length-contracted to L = L0/gamma, so the muon needs to cover less distance and survives the trip.
5. Conclude that both frames agree the muon arrives. The extra muons are evidence for special relativity because the measured flux matches the relativistic prediction and not the classical one.
6. Write the explanation in words that link the effect to the number reaching the surface, and mention the experimental observation.

Where the marks are

- A mark for naming time dilation (lifetime longer from Earth's frame) and/or length contraction (distance shorter from the muon's frame).
- A mark for linking the effect to a larger number of muons arriving at the surface than the classical prediction.
- A mark for stating that the observation supports special relativity over the classical model, not just describing the effect.

Pitfalls

- Describing dilation or contraction without saying which observer sees it, which loses the frame-specific mark.
- Stating that the muon speeds up or that its mass changes, rather than that its lifetime or distance changes.
- Explaining only that muons are 'fast' without naming the classical expectation they are compared with.

- Writing that the muons reach the ground because of relativity without linking the effect to the observed count.
- Treating the result as a proof of relativity rather than evidence consistent with it.

## Pattern 6: General relativity: GPS clocks, black holes, lensing

Syllabus: S26-U3-SHE-03 / S27-U3-SHE-03

Frequency: 1 of 10 exam years; 2 marks in all.

Marks reward: Qualitative explanation 2

Past questions: 2016R-11(b)

Summary: This point asks students to explain general relativity's predictions in context: why GPS clocks need correcting, what makes a black hole, how gravitational waves are detected and how lensing happens. Past examples are sparse (one 2-mark EXPLAIN part), so questions usually appear as a short written explanation or a qualitative comparison rather than a calculation.

Recognise: Look for a phenomenon described in words (a satellite clock, a region where light cannot escape, a detector with perpendicular arms, light from a distant star bent around a galaxy) and a verb such as explain, describe, or account for. Usually no numbers are needed beyond recalling the idea.

Steps

1. Name the effect the question is about: gravitational time dilation, black hole, gravitational wave or lensing.
2. Link it to curved spacetime: mass and energy curve spacetime, and objects and light follow the curved paths, rather than being pulled by a force as in Newtonian gravity.
3. For GPS clocks, compare two effects: weaker gravity at satellite height makes the clock run faster, while the satellite's orbital speed makes it run slower (special relativity). The net effect is a fast drift that must be corrected.
4. For black holes, use the Schwarzschild radius r_s = 2GM/c^2: inside this radius the spacetime curvature is so strong that nothing, including light, can escape.
5. For gravitational waves, state that they stretch and squeeze space transverse to their direction of travel. An interferometer arm's length change depends on its orientation: maximum when one arm is stretched while the perpendicular arm is squeezed, and little or none when both arms are affected equally.
6. For lensing, explain that light follows the curved path of spacetime around a massive object, so the observed position of a source is shifted or several images appear.

Where the marks are

- Full marks go to an explanation that names the mechanism (spacetime curvature) and links it to the observed effect.
- A mark is typically given for correctly stating the comparison, for example which effect dominates or which arm is lengthened or shortened.
- Describing the wave's effect on arms relative to direction, not just saying it 'changes distance', earns the orientation mark.
- A clear conclusion in the final sentence connecting cause to observation is often required for the second mark.

Pitfalls

- Describing gravity as a force pulling light or clocks directly, without reference to curved spacetime.
- Forgetting that GPS needs both the gravitational and the velocity effects; stating only one of them.
- Saying a gravitational wave shortens or lengthens the path along its direction of travel, when the stretch and squeeze is transverse.
- Confusing a black hole's event horizon with a physical surface, or saying matter 'stops' there rather than being unable to escape.

- Giving an answer about one arm only, without comparing how the two perpendicular arms respond.